Chemistry CSCA January 2026

Instructions

Each question has four options among which only one is correct.

Atomic weights: O: 16, Na: 23, Cu: 64, H: 1, C: 12, He: 2

Questions

Question 1

Which of the following is a chemical property of a substance?

(A) Iron nails easily rust in a humid environment

(B) The density of copper is $8.9\ g \cdot cm^{-3}$

(C) Platinum wire turns red when heated

(D) Aluminum is a silver-white metal

Question 2

Which of the following instruments is used for accurately measuring liquid volume?

(A) Measuring cylinder

(B) Beaker

(C) Balance

(D) Gas collection bottle

Question 3

In the laboratory, hydrogen is prepared using:

(A) Iron and concentrated sulfuric acid

(B) Magnesium and concentrated nitric acid

(C) Copper and dilute sulfuric acid

(D) Zinc and dilute sulfuric acid

Question 4

The structural formula of acetic acid is:

(A) CH_3CH_2OH

(B) CH_3OCH_3

(C) CH_3COOH

(D) CH_3CHO

Question 5

Which of the following statements about ^{13}C is incorrect?

(A) ^{13}C and ^{12}C have the same number of electrons

(B) ^{13}C and ^{12}C have the same number of protons

(C) ^{13}C and ^{12}C have the same number of neutrons

(D) ^{13}C is an isotope of ^{12}C

Question 6

At 25°C, which of the following solutions has a pH of 1?

(A) 0.1 mol · L^{-1} hydrochloric acid solution

(B) 0.1 mol · L^{-1} sulfuric acid solution

(C) 0.1 mol · L^{-1} sodium hydroxide solution

(D) 0.1 mol · L^{-1} acetic acid solution

Question 7

When observing the flame test for potassium, blue cobalt glass is used to:

(A) Filter out the yellow light of sodium

(B) Protect the eyes from burns

(C) Enhance the purple light of potassium

(D) Increase the observation temperature

Question 8

Which of the following substances can form a solution with water?

(A) Ethyl acetate

(B) Ethanol

(C) Benzene

(D) Calcium carbonate

Question 9

Which of the following substances is a pure substance?

(A) Medical alcohol

(B) Concrete

(C) Copper

(D) Saline solution

Question 10

Which of the following substances is a homolog of C_9H_{16}?

(A) C_4H_8

(B) C_4H_{10}

(C) C_5H_8

(D) C_8H_{10}

Question 11

In which of the following reactions does sulfuric acid act as an oxidizing agent?

(A) $2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$

(B) $H_2SO_4 + Ba(NO_3)_2 \rightarrow 2HNO_3 + BaSO_4 \downarrow$

(C) $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$

(D) $2KOH + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O$

Question 12

Which of the following substances belongs to electrolyte?

(A) Sucrose

(B) Carbon dioxide

(C) Metal aluminum

(D) NaCl

Question 13

When 1 mole of hydrogen burns completely in oxygen, the mass of water produced is:

(A) 4.5 g

(B) 9 g

(C) 27 g

(D) 18 g

Question 14

Which of the following substances is an electrolyte?

(A) Magnesium

(B) Sodium chloride

(C) Toluene

(D) Helium

Question 15

Which of the following atoms is most likely to lose electrons and form a cation?

(A) Na

(B) Mg

(C) Cl

(D) F

Question 16

Which of the following groups consists of allotropes of the same element?

(A) CH_4 and C_3H_8

(B) ^{12}C and ^{14}C

(C) CO and CO_2

(D) Diamond and graphite

Question 17

In which of the following substances is the oxidation number of Cl +5?

(A) $NaClO_3$

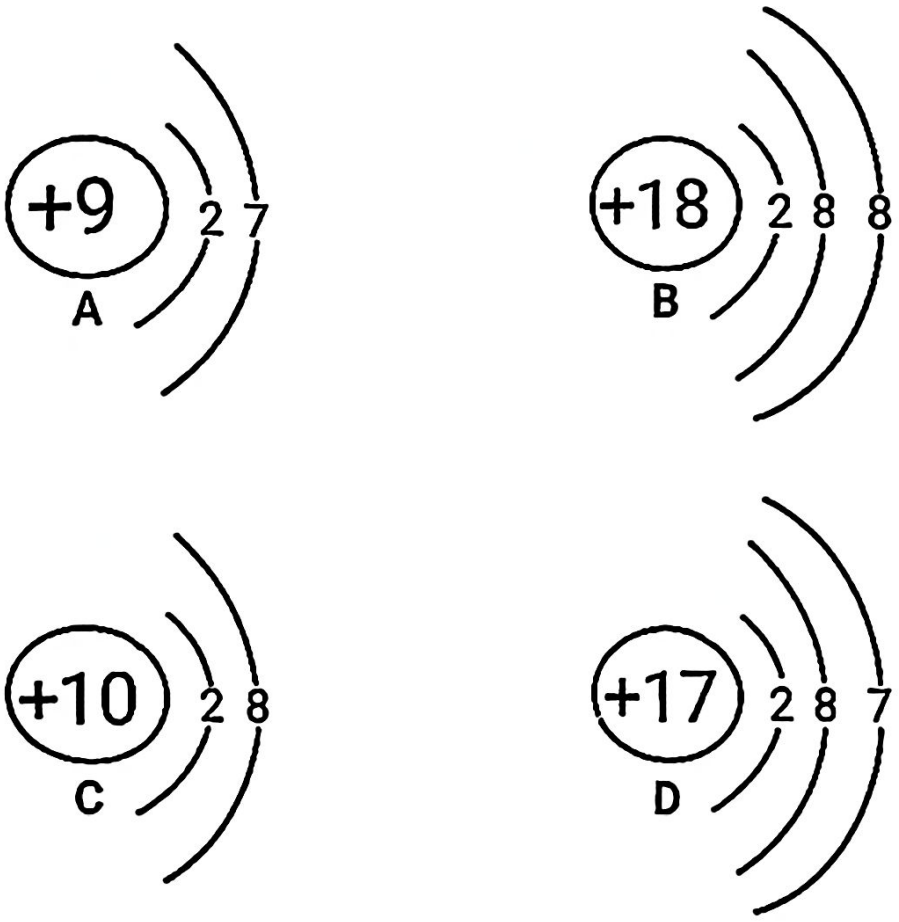
(B) $NaClO_4$

(C) $CaCl_2$

(D) NaClO

Question 18

Which of the following represents the electron configuration diagram of an argon atom?



Question 19

Which of the following gases can be collected using both downward displacement of air and water displacement?

(A) CO_2

(B) N_2

(C) H_2

(D) NH_3

Question 20

Take 50 mL of NaOH solution from 1 L of 1 $mol \cdot L^{-1}$ NaOH solution. The molar concentration of this 50 mL NaOH solution is:

(A) 0.02 $mol \cdot L^{-1}$

(B) 1 $mol \cdot L^{-1}$

(C) 2 $mol \cdot L^{-1}$

(D) 0.2 $mol \cdot L^{-1}$

Question 21

Which of the following substances can oxidize Fe^{2+} to Fe^{3+}?

(A) Cl_2

(B) Fe

(C) Zn

(D) NaOH

Question 22

At 25°C, which of the following substances can react with both HCl and NaOH?

(A) $NaHCO_3$

(B) $Ca(OH)_2$

(C) CH_3COOH

(D) NaCl

Question 23

Which of the following elements has the smallest atomic radius?

(A) O

(B) C

(C) Na

(D) F

Question 24

When the reaction reaches equilibrium, what is always equal in the reaction system?

(A) Forward reaction rate and reverse reaction rate

(B) Concentration of reactants and products

(C) The time that atoms exist in reactant molecules and product molecules

(D) Mass of reactants and products

Question 25

If 4.0 g of copper(II) oxide is completely reduced by H_2, the volume of H_2 consumed under standard conditions is:

(A) 11.2 L

(B) 22.4 L

(C) 2.24 L

(D) 1.12 L

Question 26

The English name for $NaNO_2$ is:

(A) Ammonium nitrate

(B) Sodium nitride

(C) Sodium nitrite

(D) Sodium nitrate

Question 27

Which of the following substances, when dissolved in water, produces the solution with the highest pH?

(A) NaCl

(B) HNO_3

(C) H_2S

(D) NH_3

Question 28

The reaction between hydrochloric acid and sodium hydroxide belongs to:

(A) Decomposition reaction

(B) Displacement reaction

(C) Combination reaction

(D) Neutralization reaction

Question 29

Which of the following substances contains both ionic and covalent bonds

(A) Na_2O_2

(B) CO_2

(C) H_2O

(D) KCl

Question 30

Which of the following is a displacement reaction?

(A) $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2 \uparrow$

(B) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

(C) $2H_2 + O_2 = 2H_2O$

(D) $CH_4 + 2O_2 = CO_2 + 2H_2O$

Question 31

In the laboratory preparation of oxygen from H_2O_2, the commonly used catalyst is:

(A) Manganese dioxide

(B) Potassium chloride

(C) Sodium hydroxide

(D) Potassium chlorate

Question 32

Which of the following substances has bleaching properties?

(A) $KClO_3$

(B) Dry C

(C) NaCl

(D) HClO

Question 33

In the periodic table, the element located in the second period and Group VIIA is:

(A) Oxygen

(B) Nitrogen

(C) Fluorine

(D) Chlorine

Question 34

Which of the following ions can coexist in large quantities in an aqueous solution?

(A) $Mg^{2+}, Cl^-, NH_4^+, SO_4^{2-}$

(B) $Ca^{2+}, SO_4^{2-}, CO_3^{2-}, Na^+$

(C) $CO_3^{2-}, Na^+, SO_4^{2-}, H^+$

(D) OH^- Ca^{2+}, Mg^{2+}, NO_3^-

Question 35

In a beaker containing 50 mL of saturated potassium nitrate solution, the method to change the mass fraction of the solute is:

(A) Evaporate some solvent at constant temperature

(B) Increase the solution temperature

(C) Decrease the solution temperature

(D) Add potassium nitrate solid

Question 36

What reagents can be used to detect Cl^- ions?

(A) $Ba(NO_3)_2$ and dilute nitric acid

(B) $AgNO_3$ and dilute sulfuric acid

(C) $BaCl_2$ and dilute hydrochloric acid

(D) $AgNO_3$ and dilute nitric acid

Question 37

Which of the following substances contains the largest number of atoms?

(A) 11.2 L of CH_4 under standard conditions

(B) 2 g He

(C) 3.01×10^{23} Al atoms

(D) 1 mol HF

Ouestion 38

For the endothermic reaction $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$, what can cause the equilibrium to shift in the opposite direction?

(A) Increase temperature, increase pressure

(B) Decrease temperature, decrease pressure

(C) Decrease temperature, increase pressure

(D) Increase temperature, decrease pressure

Question 39

H_2 is mixed with small amounts of CO_2 and NH_3. To purify H_2, the mixed gas should be passed through ( ) in sequence.

(A) Concentrated H_2SO_4, NaOH(

(B) CaO, NaOH(aq)

(C) NaOH(aq), CaO

(D) NaOH(aq), Concentrated H_2SO_4

Question 40

When 12 g of carbon powder burns in 0.8 mol of oxygen, and all the carbon is consumed, the resulting gas contains:

(A) Possibl / both CO and CO_2

(B) Only CO

(C) CO_2 and O_2

(D) Only CO_2

Question 41

The method to remove a small amount of $NaHCO_3$ from Na_2CO_3 solid is to:

(A) Introduce carbon dioxide

(B) Add sodium hydroxide

(C) Heat

(D) Add dilute hydrochloric acid

Question 42

For the reaction $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ (forward reaction is exothermic), which of the following will shift the equilibrium to the right.

(A) Raise the temperature

(B) Add a catalyst

(C) Increase the system pressure

(D) Remove H_2

Question 43

Regarding the reaction:

$$CH_3COOH + C_2H_5OH \xrightarrow[\Delta]{H_2SO_4} CH_3COOC_2H_5 + H_2O$$

(where concentrated sulfuric acid is added), which of the following statements is incorrect?

(A) Concentrated sulfuric acid can absorb the produced water, facilitating the reaction

(B) Concentrated sulfuric acid acts as a catalyst in the reaction

(C) This reaction can be controlled to completely convert reactants into products

(D) This reaction is an esterification reaction

Question 44

When 0.1 mol of Na reacts with 37.8 g of water, the mass fraction of the resulting NaOH solution is:

(A) 15%

(B) 10%

(C) 25%

(D) 20%

Question 45

The electron configuration of a ground-state potassium atom is:

(A) $1s^2 2s^2 2p^7 3s^2 3p^6$

(B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

(C) $1s^2 2s^2 2p^6 3s^2 3p^6 3s^2 4p^1$

(D) $1s^2 2s^1 2p^4 3s^2 3p^6 4s^2$

Question 46

A certain white powder is composed of two substances, and the following experiments

are conducted to identify its composition:

- Take a small amount of sample and add sufficient water; the solid partly dissolves. Then add a sufficient amount of dilute hydrochloric acid, bubbles are formed, and the solid completely dissolves.

  Take a small amount of sample and add sufficient dilute sulfuric acid, bubbles are produced. After shaking, there is still solid present.

The white powder may be:

(A) $CaCO_3$, $BaCO_3$

(B) $AgNO_3$, Na_2CO_3

(C) $NaOH$, $CuSO_4$

(D) Na_2CO_3, $BaCO_3$

Question 47

Scientists studied a new material containing three short-period elements X, Y, and Z with increasing nuclear charges. The outermost electron number of X is twice that of its sub-outer layer. The M-shell electron number of Y is half of its L-shell electron number. The sub-outer electron number of Z is one more than its outermost electron number. Which of the following statements is incorrect?

(A) The compound of X and Z is insoluble in water

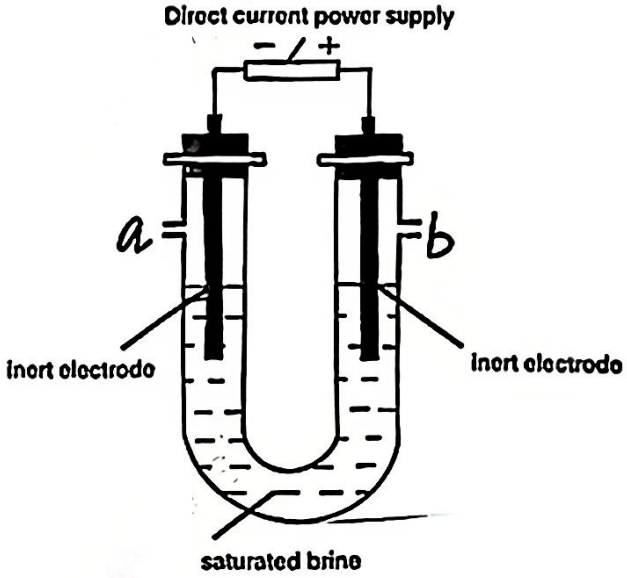
(B) All three elements are nonmetals

(C) X and Y belong to the same group

(D) The atomic radius of Y is smaller than that of Z

Question 48

The laboratory electrolysis of brine is shown in the diagram (see original image: direct current (DC) power supply, inert electrodes, saturated brine). Which of the following statements is correct?



(A) After adding phenolphthalein to the U-tube, the solution near port a turns red over time

(B) The gas released at port a turns moist litmus paper red

(C) The gas released at port b burns with a blue flame

(D) For every 1 mole of electrons transferred in the circuit, the volume of gas released at port b is 22.4 L (standard conditions)